

JVD Management System - Complete Architecture & Structure

Internal Paperless Operations Platform

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1. System Overview

The JVD Management System is an internal-only, paperless operations platform for JVD Events and Travels Management Co. It replaces every manually signed and filled paper with a digital, trackable, and auditable workflow.

What the system manages:

- Bus rental operations (field trips, events, corporate transport)
- Flight logistics and travel coordination
- Passport and visa processing services
- Fleet maintenance and bus accreditation
- Procurement, inventory, and supplier management
- All internal paperwork: Work Orders, Job Orders, Purchase Orders, Contracts

What the system is NOT:

- Not a customer-facing booking portal
- Not an e-commerce or payment platform
- Not a public website

2. Account System & Authentication

2.1 How Accounts Work

Accounts are created exclusively by the Admin (CEO or designated Super Admin). There is no

self-registration - this is a closed corporate system.

Account Lifecycle:

```
+-----+
| Admin creates account |
|   -> Assigns role (Employee, HR, Accounting, Agent, etc) |
|   -> Sets department assignment |
|   -> System generates temporary password |
|   -> Employee receives credentials securely |
| |
| First Login |
|   -> Employee forced to change temporary password |
|   -> Employee MUST set up Google Authenticator (2FA) |
|   -> Account becomes fully active |
| |
| Ongoing |
|   -> Every login requires password + 2FA TOTP code |
|   -> Sessions auto-expire after inactivity |
|   -> Admin can deactivate/revoke any account instantly |
+-----+
```

2.2 Authentication Flow

```
User enters Email + Password
|
v
+-----+      NO
| Credentials |-----> "Invalid credentials" (generic error)
| valid?      |
+-----+
| YES
v
Enter Google Authenticator 6-digit TOTP code
|
v
+-----+      NO
| TOTP code  |-----> "Invalid code" + rate limit after 5 attempts
| valid?      |
+-----+
| YES
v
+-----+      NO
| Account    |-----> "Account suspended. Contact Admin."
| active?    |
+-----+
| YES
v
Session created -> Redirected to role-based dashboard
```

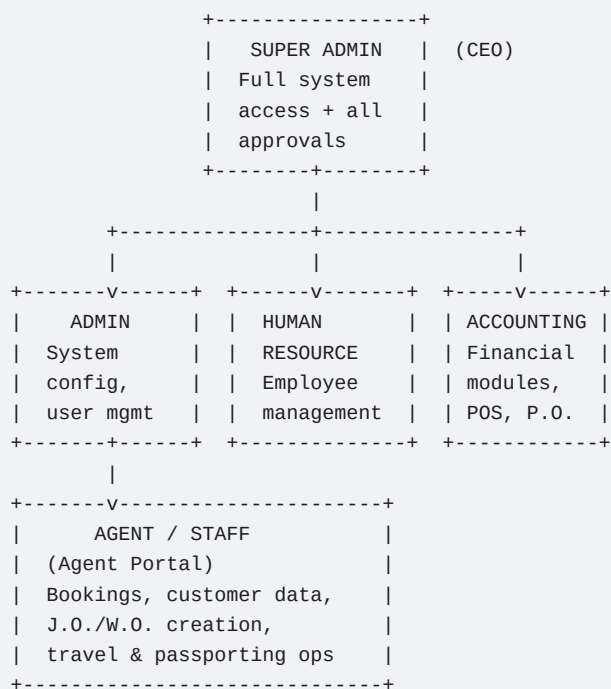
2.3 Account Fields

Field	Description
employee_id	Unique company-assigned ID
email	Company email (login credential)
password	Bcrypt/Argon2 hashed
first_name / last_name	Employee identity
role	One of the defined roles (see Section 3)
department	Department assignment
2fa_secret	Google Authenticator TOTP secret
is_active	Admin can enable/disable
last_login	Timestamp for audit
created_by	Which admin created this account

3. Roles, Hierarchy & Access Matrix

3.1 Role Hierarchy

Based on the whiteboard diagram from the CEO meeting, the system has these distinct roles:



3.2 Access Matrix

Each [Y] means the role can view and interact with that module. Each [LOCK] means approval/override authority.

Module	Super Admin (CEO Admin		Human Resource	Accounting	Agent / Staff
Dashboard (Overview)	[LOCK] Full KPIs	[Y] System stats	[Y] HR stats	[Y] Financial sta	[Y] My tasks
User/Account Management	[LOCK] Create/Edi	[Y] Create/Edit	[X]	[X]	[X]
Accounting / POS	[LOCK] View + App	[X]	[X]	[Y] Full CRUD	[X]
Procurement	[LOCK] View + App	[Y] View	[X]	[Y] Financial sid	[Y] Create reques
Purchase Orders (P.O.)	[LOCK] Final appr	[Y] View all	[X]	[Y] Verify + Proc	[Y] Create/Draft
Job Orders (J.O.)	[LOCK] View all	[Y] Manage	[X]	[Y] View (cost tr	[Y] Create + Upda
Work Orders (W.O.)	[LOCK] View all	[Y] Manage	[X]	[Y] View (cost tr	[Y] Create + Upda
Accreditations	[LOCK] View + App	[Y] Manage	[X]	[X]	[Y] View assigned
Inventory & Supplies	[LOCK] View all	[Y] Full CRUD	[X]	[Y] View (cost)	[Y] View/Request
PMS (Predictive Maintena	[LOCK] View all	[Y] Full CRUD	[X]	[X]	[Y] Log entries
Bus Management & Accredi	[LOCK] View all	[Y] Full CRUD	[X]	[X]	[Y] View schedule
Travel Division	[LOCK] View all	[Y] View	[X]	[X]	[Y] Full operatio
Passporting & Visa Proce	[LOCK] View all	[Y] View	[X]	[X]	[Y] Full operatio
Customer/Passenger Data	[LOCK] View all	[Y] View all	[X]	[X]	[Y] CRUD (own cli
Legal Documents & Contra	[LOCK] View + Sig	[Y] View all	[X]	[Y] View financia	[Y] Upload/Draft
HR / Employee Management	[LOCK] Full acces	[Y] View	[Y] Full CRUD	[X]	[X]
Audit Logs	[LOCK] Full acces	[Y] View	[X]	[X]	[X]
System Settings	[LOCK] Full acces	[Y] Limited	[X]	[X]	[X]

4. Department Modules

4.1 Accounting Department

Purpose: Centralized financial management for the entire company.

Sub-Module	Description
------------	-------------

POS (Point of Sale)	Record and track all sales transactions for bus rentals, travel packages, and s
Billing & Invoicing	Generate digital invoices for clients. Track payment status (Unpaid -> Partial
Financial Reports	Revenue summaries, expense tracking, profit/loss statements filtered by date, d
P.O. Financial Verification	Accounting verifies the budget and financial validity of Purchase Orders before

4.2 Procurement Department

Purpose: Manage everything the company needs to acquire and get certified.

Sub-Module	Description
Accreditation Management	Track and manage company accreditations (LTFRB, DOT, etc.) and individual drive
Purchase Order (P.O.) Management	Create, track, and route P.O.s through the approval pipeline. Link P.O.s to sup
Job Order (J.O.) Management	Create and manage job orders - the primary record of each service engagement (b
Work Order (W.O.) Management	Create and manage work orders for internal tasks - bus repairs, office maintena
Supplier Management	Maintain a database of suppliers (fuel, parts, food, etc.) with contact info, p

4.3 Inventory Department

Purpose: Track all physical assets and supplies the company owns.

Sub-Module	Description
Supplies Tracking	Current stock levels of fuel, spare parts, office supplies, event materials. Lo
PMS (Predictive Maintenance System)	Schedule and track bus maintenance based on mileage, usage hours, and inspectio
Bus Fleet Registry	Every bus in the fleet with: plate number, model, capacity, current condition,
Bus Accreditation	Track regulatory compliance for each bus unit (e.g., LTFRB franchise validity,

4.4 Travel Division

Purpose: Manage all travel-related service operations.

Sub-Module		Description
Passporting		Track passport application processing for clients: document submission -> proce
Visa Processing		Track visa applications: requirements checklist -> submission -> embassy status
Customer/Passenger Management	Data	Client profiles containing: Full Name, Birth Date, Passport Number, Contact Inf
Passenger Checklist		Per-passenger document checklist (valid ID, passport copy, visa copy, medical c
Data Sheets		Travel manifests and passenger lists for each trip. Auto-generated from J.O. pa
Legal Documents & Contracts		Digital storage for service agreements, liability waivers, and contracts. Linke

5. Paperless Document Lifecycle Flows

5.1 Purchase Order (P.O.) Flow

This is the most critical approval pipeline - it replaces the paper P.O. that currently gets physically signed by multiple people.

```
Agent/Staff creates P.O. Draft
(selects supplier, items, quantities, estimated cost)
  |
  v
Status: DRAFT
-> Staff reviews and submits
  |
  v
Status: PENDING ACCOUNTING REVIEW
-> Accounting receives notification
-> Accounting verifies budget, pricing, supplier validity
  |
  |-- REJECTED -> Returns to Staff with notes -> Staff revises -> resubmit
  |
  v
Status: PENDING CEO APPROVAL
-> CEO (Super Admin) receives notification
-> CEO reviews with Accounting's verification notes
  |
  |-- REJECTED -> Returns with CEO notes -> cycle restarts
  |
  v
Status: APPROVED [Y]
-> P.O. number generated (auto-incrementing)
-> Digital "signature" timestamp recorded (who approved, when, IP)
-> Inventory auto-updated when items are received
-> PDF version auto-generated for archival/printing if needed
```

5.2 Job Order (J.O.) Flow

The J.O. is the master record of a service engagement - when JVD is hired for a bus trip, event, or travel package.

```
Agent/Staff creates J.O.  
(client info, service type, date, destination, bus assignment, pricing)  
  |  
  v  
Status: CREATED  
-> Link customer/passenger data  
-> Attach passenger checklist + data sheet  
-> Assign bus unit(s) from fleet  
-> Assign driver(s)  
  |  
  v  
Status: CONFIRMED  
-> Bus schedule updated (calendar blocked)  
-> PMS checks if assigned bus needs maintenance before trip date  
-> If maintenance needed -> auto-generates W.O. (see 5.3)  
  |  
  v  
Status: IN PROGRESS  
-> Trip/event is underway  
  |  
  v  
Status: COMPLETED  
-> Accounting generates invoice from J.O. data  
-> All documents (contract, manifest, checklist) archived digitally  
-> J.O. marked complete with timestamp
```

5.3 Work Order (W.O.) Flow

The W.O. is for internal operational tasks - primarily bus maintenance and repairs.

```
W.O. created (manually by staff OR auto-generated by PMS)  
(bus unit, task description, parts needed, urgency level)  
  |  
  v  
Status: OPEN  
-> Assigned to maintenance staff/driver  
-> If parts needed -> links to P.O. (triggers P.O. flow if parts must be purchased)  
  |  
  v  
Status: IN PROGRESS  
-> Staff logs work done, parts used, hours spent  
  |  
  v  
Status: COMPLETED  
-> Bus status updated in Fleet Registry  
-> PMS recalculates next maintenance window  
-> Cost logged for Accounting
```

5.4 Passporting & Visa Processing Flow


```
Staff creates Passport/Visa case
(client info, type of service, target country if visa)
  |
  v
Status: REQUIREMENTS GATHERING
-> System shows required documents checklist
-> Staff checks off documents as client submits them
-> Missing items flagged with reminders
  |
  v
Status: DOCUMENTS COMPLETE
-> All checklist items verified
  |
  v
Status: SUBMITTED FOR PROCESSING
-> Submitted to DFA (passport) or Embassy (visa)
-> Staff records submission date and reference number
  |
  v
Status: PROCESSING
-> Staff monitors and updates status periodically
  |
  |-- DENIED (visa) -> Record reason -> Notify client options
  |
  v
Status: READY FOR RELEASE
-> Passport/Visa received
-> Staff notifies client for pickup/delivery
  |
  v
Status: RELEASED [Y]
-> Case closed and archived
```

5.5 Predictive Maintenance System (PMS) Flow

```
System continuously monitors per bus:
|-- Total mileage since last service
|-- Days since last inspection
|-- Reported issues from drivers (logged via W.O.)
|-- Regulatory inspection due dates
    |
    v
+-----+
| Threshold reached?          |
| (e.g., 10,000 km since last |
| oil change, or 90 days since |
| last inspection)            |
+-----+
    | YES
    v
System auto-generates a W.O. with:
-> Bus unit details
-> Recommended maintenance tasks
-> Priority level (Routine / Urgent / Critical)
-> Alerts Admin + assigned driver
    |
    v
W.O. enters normal Work Order flow (see 5.3)
```

6. Data Models & Relationships

6.1 Entity Relationship Overview

erDiagram

USER ||--o{ AUDIT_LOG : "actions logged"

USER {

```

    int id PK
    string employee_id UK
    string email UK
    string password_hash
    string first_name
    string last_name
    enum role
    string department
    string totp_secret
    boolean is_active
    datetime last_login
    int created_by FK

```

}

CUSTOMER ||--o{ PASSENGER : "has passengers"

CUSTOMER ||--o{ JOB_ORDER : "associated with"

CUSTOMER {

```

    int id PK
    string first_name
    string last_name
    string email
    string phone
    string address
    text notes

```

}

PASSENGER {

```

    int id PK
    int customer_id FK
    string first_name
    string last_name
    date birth_date
    string passport_no
    string contact_no
    jsonb checklist_status

```

}

BUS ||--o{ JOB_ORDER : "assigned to"

BUS ||--o{ WORK_ORDER : "maintained by"

BUS ||--o{ BUS_ACCREDITATION : "certified by"

BUS {

```

    int id PK
    string plate_number UK
    string model
    int seating_capacity
    enum status
    float total_mileage
    date last_service_date
    date next_service_due
    int assigned_driver FK

```

}

JOB_ORDER ||--o{ PASSENGER : "includes"

JOB_ORDER ||--o{ LEGAL_DOCUMENT : "attached"

JOB_ORDER {

```

    int id PK

```

```
    string jo_number UK
    int customer_id FK
    int bus_id FK
    int created_by FK
    enum service_type
    enum status
    date service_date
    string destination
    decimal total_cost
    text notes
}

WORK_ORDER {
    int id PK
    string wo_number UK
    int bus_id FK
    int assigned_to FK
    int created_by FK
    enum status
    enum priority
    text description
    text parts_used
    decimal cost
    boolean auto_generated
}

PURCHASE_ORDER ||--o{ PO_LINE_ITEM : "contains"
PURCHASE_ORDER {
    int id PK
    string po_number UK
    int supplier_id FK
    int created_by FK
    int verified_by FK
    int approved_by FK
    enum status
    decimal total_amount
    text rejection_notes
    datetime approved_at
}

SUPPLIER ||--o{ PURCHASE_ORDER : "fulfills"
SUPPLIER {
    int id PK
    string company_name
    string contact_person
    string phone
    string email
    text address
}

PASSPORT_CASE {
    int id PK
    int customer_id FK
    int passenger_id FK
    int handled_by FK
    enum case_type
    enum status
    jsonb checklist
    string reference_number
    date submitted_date
```

```
    date release_date
}

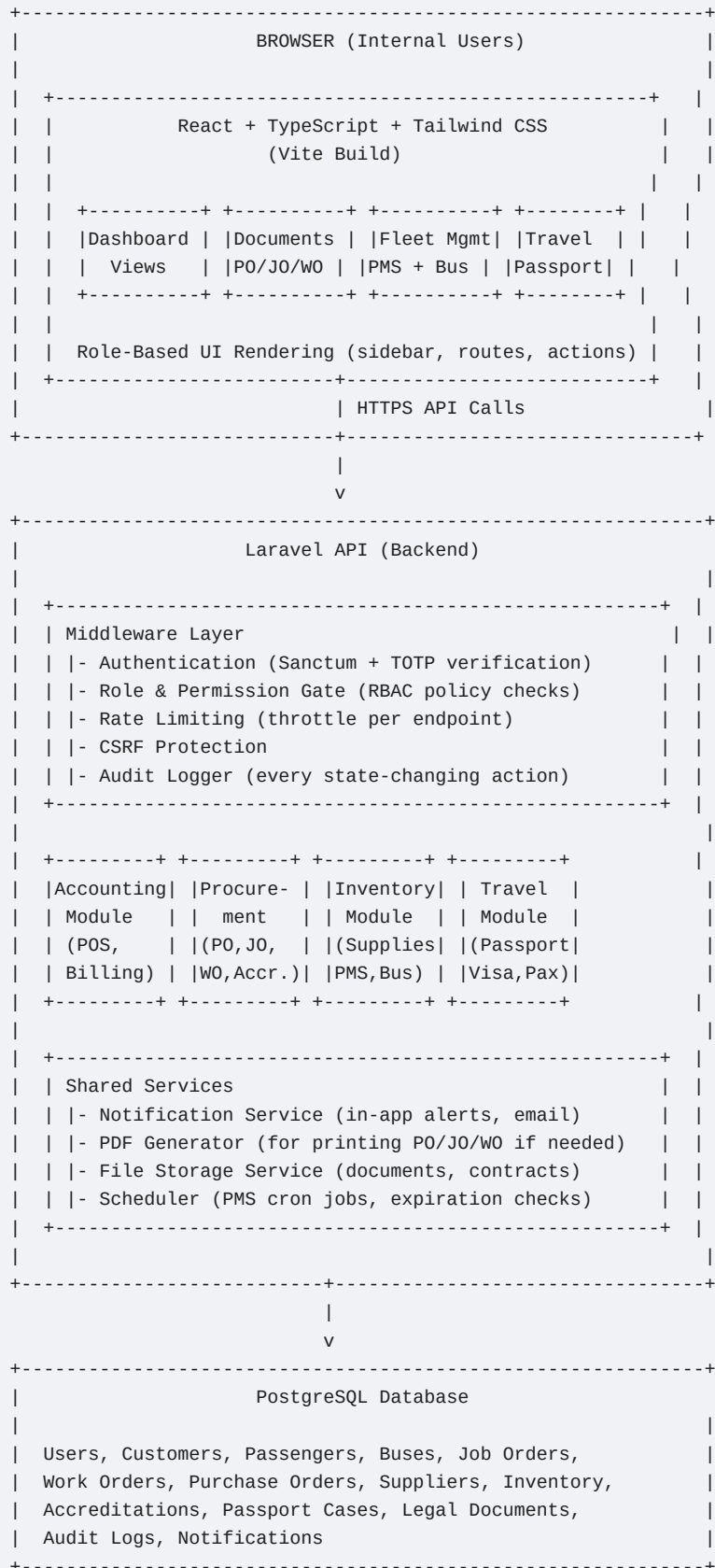
ACCREDITATION {
    int id PK
    enum entity_type
    int entity_id
    string accreditation_type
    string issuing_body
    date issue_date
    date expiry_date
    enum status
    string document_url
}

INVENTORY_ITEM {
    int id PK
    string item_name
    string category
    int quantity
    int reorder_level
    string unit
    decimal unit_cost
}

AUDIT_LOG {
    int id PK
    int user_id FK
    string action
    string module
    string entity_type
    int entity_id
    jsonb old_values
    jsonb new_values
    string ip_address
    datetime created_at
}
```

7. System Architecture Diagram

7.1 High-Level Architecture



7.2 Frontend Route Structure

/login	-> Login page (public, unauthenticated)
/setup-2fa	-> First-time 2FA setup (post first-login)
/dashboard	-> Role-specific overview dashboard
/accounting	
/accounting/pos	-> Point of Sale transactions
/accounting/billing	-> Invoices & payment tracking
/accounting/reports	-> Financial reports & summaries
/procurement	
/procurement/purchase-orders	-> P.O. list, create, approval pipeline
/procurement/job-orders	-> J.O. list, create, manage
/procurement/work-orders	-> W.O. list, create, manage
/procurement/accreditations	-> Company & driver accreditation registry
/procurement/suppliers	-> Supplier database
/inventory	
/inventory/supplies	-> Stock levels & reorder alerts
/inventory/fleet	-> Bus registry & status
/inventory/bus-accreditation	-> Bus regulatory compliance tracking
/inventory/pms	-> Predictive Maintenance dashboard
/travel	
/travel/passporting	-> Passport case tracker
/travel/visa-processing	-> Visa case tracker
/travel/customers	-> Customer/passenger database
/travel/documents	-> Legal documents & contracts
/hr	
/hr/employees	-> Employee directory & management
/admin	
/admin/users	-> Account creation & management
/admin/audit-logs	-> System-wide action log
/admin/settings	-> System configuration

8. Security Architecture

8.1 Authentication Security

Layer	Implementation
Password Policy	Minimum 12 characters, must include uppercase, lowercase, number, and special c
Password Storage	Argon2id hashing (Laravel default)
2FA	Google Authenticator TOTP - mandatory for all accounts, no exceptions
Session Duration	8-hour max session, 30-minute inactivity timeout

Login Throttling	5 failed attempts -> 15-minute logout -> alert to Admin
Account Lockout	Admin-only unlock after 10 consecutive failures

8.2 Authorization Security

Layer	Implementation
RBAC Enforcement	Laravel Gates & Policies - checked on EVERY API request
UI-Level Gating	React renders sidebar items, routes, and action buttons based on role
API-Level Gating	Even if UI is bypassed, backend rejects unauthorized requests
Data Scoping	Agents only see their own clients. Accounting only sees financial data.

8.3 Data & Infrastructure Security

Layer	Implementation
HTTPS	TLS 1.3 enforced - all traffic encrypted in transit
CSRF	Laravel CSRF tokens on all state-changing endpoints
SQL Injection	Eloquent ORM parameterized queries - no raw SQL
XSS	React auto-escapes DOM output + Content Security Policy headers
File Upload	Strict MIME-type validation, virus scanning, sandboxed storage
Audit Trail	Every create, update, delete, and approval action logged with user ID, timestamp

IMPORTANT: The whiteboard images from the CEO meeting have been fully decoded and incorporated into this architecture. The "picture" referenced in the PDF contained the department structure, document types (P.O./J.O./W.O.), the role matrix (Admin, Agent Portal, Employee, HR, Accounting), and the customer data fields (Name, Birth Date, Passport No., Contact).